

A Comparative Framework for Diabetes Prediction Using Balanced Machine Learning and Deep Learning Models

1st Pakrigna Long

Department of Applied Mathematics
and Statistics
Institute of Technology of Cambodia
Phnom Penh, Cambodia
long.pakrigna@itc.edu.kh

2nd Botumraksa Choub

Department of Applied Mathematics
and Statistics
Institute of Technology of Cambodia
Phnom Penh, Cambodia
botum.raksa@gmail.com

3rd Kimlong Ngin

Institute of Information Technology
University of Heng Samrin
Thbongkhmum
Tboung Khmum, Cambodia
nginkimlong@gmail.com

Abstract—*Diabetes mellitus, one of the most common chronic diseases, requires early diagnosis to reduce complications and improve patient outcomes. Machine learning (ML) and deep learning (DL) can predict diabetes. Class imbalance remains a major prediction issue. We compare balanced machine learning and deep learning models for diabetes prediction in this study. The BRFSS 2015 diabetes dataset was used. Machine learning models used the Synthetic Minority Oversampling Technique (SMOTE) to address class imbalance, while deep learning models used the original class distribution. In this study, four machine learning models namely logistic regression (LR), random forest (RF), naïve bayes (NB), and extreme gradient boosting (XGBoost), together with three deep learning models, namely artificial neural network (ANN), convolutional neural network (CNN), and recurrent neural network (RNN) were assessed. Two feature configurations were tested: all features and AnyHealthcare removed to assess its Diabetes prediction performance. Precision, recall, accuracy, and F1-score assessed performance. In experiments, ANN had the highest accuracy of 84.99%, CNN 84.97%, and XGBoost 81.70%. Deep learning models were slightly more accurate, but XGBoost had a better precision-recall balance. A minimal performance degradation after feature removal indicated strong feature redundancy in the dataset. The framework assesses balanced ML and DL diabetes prediction methods and their model robustness and feature contribution.*

Keywords—*Diabetes prediction, machine learning, deep learning, SMOTE, feature selection.*

I. INTRODUCTION

Diabetes mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels resulting from inadequate insulin production or ineffective insulin utilization [3]. According to global health reports, diabetes continues to affect millions of individuals and contributes significantly to mortality and healthcare expenditures [1]. Early detection of diabetes is therefore critical for effective disease management and prevention of severe complications.

Recent advancements in artificial intelligence have enabled healthcare practitioners to leverage machine learning and deep learning techniques for disease prediction [7], [11]. These techniques can analyze large volumes of health-related data and identify hidden patterns associated with diabetes risk. However, healthcare datasets frequently suffer from class imbalance, where the number of healthy individuals significantly exceeds the number of diabetic cases. Such imbalance often leads to biased predictive models.

To address this issue, data balancing techniques such as the Synthetic Minority Oversampling Technique (SMOTE) have been widely adopted [4]. Among the available predictors, healthcare accessibility represented by the AnyHealthcare attribute may influence diabetes prediction performance. However, its contribution remains unclear. Therefore, this study additionally investigates the effect of removing the AnyHealthcare feature to evaluate the robustness of machine learning and deep learning models under feature reduction conditions. The objective of this work is not to introduce a new predictive model or optimization technique. Instead, the study focuses on providing an empirical comparison of commonly used machine learning and deep learning methods under a unified experimental setting. Such comparative evaluations are valuable for identifying practical trade-offs between model complexity, predictive performance, and feature robustness in diabetes prediction tasks.

This study presents a comparative evaluation framework for assessing machine learning and deep learning approaches for diabetes prediction under consistent preprocessing, class-balancing, and feature-reduction settings:

- Using all available features.
- Using a feature-removal strategy.

Although previous studies have compared machine learning and deep learning methods for diabetes prediction, differences in datasets, preprocessing procedures, class-balancing strategies, and evaluation metrics make direct comparison difficult. Therefore, this study provides a unified experimental evaluation of selected machine learning and deep learning models on the BRFSS 2015 diabetes dataset under consistent preprocessing and evaluation settings. In addition, the effect of removing the AnyHealthcare feature is investigated to examine model robustness and feature sensitivity.

II. LITERATURE REVIEW

Diabetes prediction has become a prominent application area of machine learning and deep learning in healthcare. Traditional machine learning algorithms such as Logistic Regression (LR), Naïve Bayes (NB), Random Forest (RF), and Support Vector Machine (SVM) have been widely employed due to their ability to classify diabetic and non-diabetic patients using clinical and demographic data. Among these methods, ensemble-based models such as Random Forest have demonstrated strong predictive

performance by effectively handling nonlinear relationships and reducing overfitting.

In addition, advanced ensemble techniques such as Extreme Gradient Boosting (XGBoost) have gained significant attention because of their superior classification accuracy and robustness when dealing with large healthcare datasets. XGBoost utilizes gradient boosting principles to improve prediction performance through iterative optimization of weak learners [5].

Deep learning models have also been increasingly adopted for diabetes prediction. Artificial Neural Networks (ANNs) can learn complex patterns from health indicators, while Convolutional Neural Networks (CNNs) have shown effectiveness in extracting high-level feature representations. Recurrent Neural Networks (RNNs), although primarily designed for sequential data, have been explored for medical prediction tasks due to their capability to capture feature dependencies [6], [9].

Most previous studies focused solely on maximizing prediction accuracy and rarely examined whether specific healthcare-related attributes significantly contribute to predictive performance. Feature sensitivity analysis remains underexplored in diabetes prediction research [10]. Techniques such as the Synthetic Minority Oversampling Technique (SMOTE) have therefore been introduced to balance class distributions and improve classification reliability. However, limited studies have simultaneously compared balanced machine learning and deep learning models while investigating the impact of feature removal on diabetes prediction performance. This study addresses this gap by providing a comprehensive comparison of seven predictive models using SMOTE-balanced data under both all-feature and feature-removal scenarios.

Recent studies have demonstrated that ensemble learning methods such as XGBoost and Random Forest frequently outperform conventional classifiers in healthcare prediction tasks due to their ability to model nonlinear interactions among risk factors [8]. Similarly, deep learning models such as ANN and CNN have shown promising performance in diabetes prediction, particularly when large datasets are available. However, comparative studies evaluating balanced machine learning models against deep learning models while investigating feature contribution remain limited.

III. METHODOLOGY

A. Research Framework

This study proposes a diabetes prediction framework using balanced machine learning and deep learning models. The workflow consists of data preprocessing, class balancing using SMOTE, model training, feature removal analysis, and performance evaluation as shown in Fig. 1. Four machine learning models (logistic regression, random forest, naïve bayes, and XGBoost) and three deep learning models (ANN, CNN, and RNN) were evaluated under two scenarios: using all features and using a reduced feature set obtained by removing the AnyHealthcare feature.

B. Dataset and Preprocessing

The study utilized the BRFSS 2015 Diabetes Health Indicators dataset obtained from the Centers for Disease

Control and Prevention (CDC) [2]. The dataset contains health-related indicators associated with diabetes risk.

The dataset consists of 253,680 records distributed among three classes: No Diabetes, Prediabetes, and Diabetes, with the majority of samples belonging to the No Diabetes category. This imbalance motivated the use of SMOTE for the machine learning models. For the predictors, the dataset contains demographic, behavioral, and clinical health indicators, including:

- HighBP: Indicates whether the individual has high blood pressure.
- HighChol: Indicates whether the individual has high cholesterol.
- CholCheck: Indicates whether cholesterol was checked within the past 5 years.
- BMI: Body Mass Index, a measure of body fat based on height and weight.
- Smoker: Indicates whether the individual has smoked at least 100 cigarettes in their lifetime.
- Stroke: Indicates whether the individual has ever been diagnosed with a stroke.
- HeartDiseaseorAttack: Indicates whether the individual has coronary heart disease or has experienced a heart attack.
- PhysActivity: Indicates whether the individual engaged in physical activity during the past 30 days.
- Fruits: Indicates whether the individual consumes fruit regularly.
- Veggies: Indicates whether the individual consumes vegetables regularly.
- HvyAlcoholConsump: Indicates whether the individual engages in heavy alcohol consumption.
- AnyHealthcare: Indicates whether the individual has access to any form of healthcare coverage.
- GenHlth: Self-reported general health status.
- MentHlth: Number of days during the past 30 days when mental health was not good.
- PhysHlth: Number of days during the past 30 days when physical health was not good.
- DiffWalk: Indicates whether the individual has difficulty walking or climbing stairs.
- Sex: Biological sex of the individual.
- Age: Age category of the individual.
- Education: Highest level of education attained.
- Income: Annual household income category.

Data preprocessing was performed to improve data quality and ensure compatibility with machine learning and deep learning algorithms.

- Feature and target separation: the target variable (Diabetes_012) was separated from the predictor variables. Diabetes_012 is the response variable provided in the BRFSS 2015 Diabetes Health Indicators dataset and contains three classes; 0 = No Diabetes, 1 = Prediabetes, and 2 = Diabetes.
- Therefore, the prediction task was formulated as a three-class classification problem.
- Feature scaling: StandardScaler was used to normalize numerical values and improve model

convergence. The standardized value is computed as:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where:

- x = original value
 - μ = mean
 - σ = standard deviation
- Train-test split: the training set was used to develop the models, while the testing set was reserved for performance evaluation. The dataset was divided into: 70% Training Set 30% Testing Set. A fixed random seed of 42 was used throughout the experiments to guarantee reproducibility and consistency of results across different runs.
 - To facilitate reproducibility, all experiments were implemented using Python 3.14 with Scikit-learn, XGBoost, and TensorFlow/Keras libraries. A fixed random seed of 42 was applied during data splitting and model training whenever supported by the implementation.

C. Data Balancing using SMOTE

Healthcare datasets frequently exhibit class imbalance, where diabetic cases are underrepresented compared to non-diabetic cases. Such imbalance can bias predictive models toward the majority class.

To address this issue, the Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training datasets of the machine learning models. The deep learning models were trained using the original imbalanced data without synthetic oversampling. SMOTE generates synthetic minority samples according to:

$$x_{new} = x_i + \lambda(x_{nm} - x_i) \quad (2)$$

where:

- x_i is a minority-class sample,
- x_{nm} is its nearest neighbor,
- λ is a random value between 0 and 1.

D. Model Implementation

Four machine learning algorithms were employed in this study: logistic regression (LR), naïve bayes (NB), random forest (RF), and extreme gradient boosting (XGBoost). Logistic regression and naïve bayes serve as traditional classification approaches based on probabilistic learning, while random forest and XGBoost represent ensemble learning techniques capable of capturing complex nonlinear relationships in the data. These models were selected due to their widespread use in healthcare analytics and their ability to provide a comprehensive comparison between conventional and advanced machine learning methods. The models were trained using the balanced dataset generated through SMOTE and evaluated using Accuracy, Precision, Recall, and F1-score.

Three deep learning architectures were evaluated, namely artificial neural network (ANN), convolutional neural network (CNN), and recurrent neural network (RNN). ANN was utilized as a baseline deep learning model

for learning nonlinear relationships among health indicators, while CNN was employed to automatically extract informative feature representations from the input data. RNN was included to investigate the capability of sequential learning structures in diabetes prediction.

The machine learning and deep learning models were implemented using different data preparation strategies. For the machine learning models, class imbalance was addressed using the Synthetic Minority Oversampling Technique (SMOTE), which generated a balanced training dataset to improve minority class representation. The machine learning models were trained using the balanced dataset. In contrast, the deep learning models were trained using the original dataset without applying SMOTE. The deep learning models have the ability to learn complex nonlinear relationships directly from large datasets. Training the deep learning models on the original data preserves the natural class distribution and avoids potential noise introduced by synthetic samples. The configurations of the evaluated models are summarized in Table I.

This design choice was made because deep learning architectures are capable of learning complex feature representations directly from large datasets and may not necessarily benefit from synthetic oversampling. Therefore, SMOTE was applied only to the machine learning models to evaluate its effectiveness in improving traditional classifier performance.

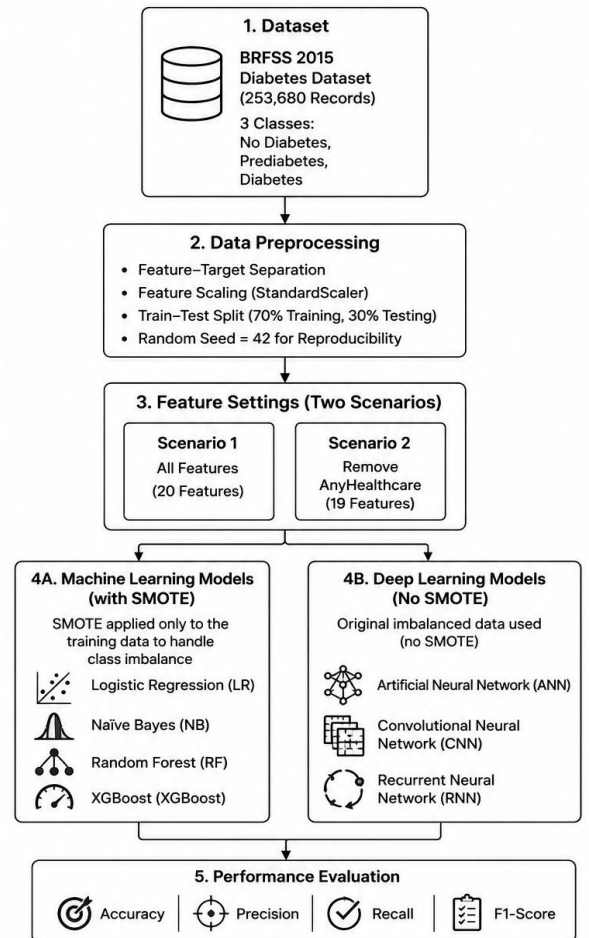


Fig. 1. Proposed diabetes prediction framework

TABLE I. MODEL CONFIGURATIONS

#	Model	Configuration
1	Logistic regression	- max_iter = 800 - multi_class = 'multinomial'
2	Naïve bayes	- Gaussian Naïve Bayes with default parameters
3	Random forest	- n_estimators = 90 - max_depth = None
4	XGBoost	- n_estimators = 110 - learning_rate = 0.13 - max_depth = 11
5	ANN	- Dense (32), activation = 'relu' - Dense (3), activation = 'softmax' - optimizer = Adam(learning_rate=0.001) - Early_Stopping (patience=10) - epochs=20, batch_size=32
6	CNN	- Conv1D (32), kernel_size=2, activation = 'relu' - Dense (3), activation = 'softmax' - optimizer = Adam(learning_rate=0.001) - Early_Stopping (patience=10) - epochs=20, batch_size = 32
7	RNN	- SimpleRNN (32), activation='tanh' - Dense (3), activation = 'softmax' - optimizer = Adam(learning_rate=0.001) - Early_Stopping (patience=10) - epochs=20, batch_size=32

IV. RESULTS AND DISCUSSION

A. Evaluation metrics

The predictive performance of all models was evaluated using four classification metrics.

- Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

- Precision

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

- Recall

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

- F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (6)$$

where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

These metrics provide a comprehensive assessment of classification performance, particularly in imbalanced healthcare datasets.

B. Results

The performance comparison of machine learning and deep learning models for diabetes prediction using two experimental settings: (i) all available features and (ii) one feature removed as shown in Table II.

The results indicate that ANN achieved the highest classification accuracy of 84.99% when all features were used, followed closely by CNN with 84.97%. Among the machine learning models, XGBoost achieved the best performance with an accuracy of 84.50% and the highest F1-score of 81.70%, indicating a strong balance between precision and recall.

The removal of the AnyHealthcare feature produced negligible changes in predictive performance across all evaluated models. For example, XGBoost accuracy changed only from 84.50% to 84.51%, Random Forest from 82.92% to 82.93%, and ANN from 84.99% to 84.95%. These results suggest that the AnyHealthcare attribute contributes limited unique information for diabetes classification and that other health indicators provide sufficient predictive capability.

The superior performance of XGBoost can be attributed to its gradient boosting mechanism, which effectively captures complex nonlinear relationships among diabetes risk factors.

TABLE II. COMPARISON OF THE MODEL PERFORMANCE ON BRFS 2015 DIABETES DATASET

#	Model	All features		One feature removal	
		Accuracy (%)	Precision, Recall and F1 Score	Accuracy (%)	Precision, Recall and F1 Score
1	Logistic regression	63.99	P: 0.8518 R: 0.6399 F1: 0.7172	64.06	P: 0.8517 R: 0.6406 F1: 0.7176
2	Naïve bayes	66.83	P: 0.8338 R: 0.6683 F1: 0.7265	66.98	P: 0.8343 R: 0.6698 F1: 0.7287
3	Random forest	82.92	P: 0.8025 R: 0.8292 F1: 0.8145	82.93	P: 0.8026 R: 0.8293 F1: 0.8146
4	XGBoost	84.50	P: 0.8038 R: 0.8450 F1: 0.8170	84.51	P: 0.8038 R: 0.8451 F1: 0.8170
5	ANN	84.99	P: 0.8044 R: 0.8499 F1: 0.8081	84.95	P: 0.8049 R: 0.8495 F1: 0.8059
6	CNN	84.97	P: 0.8043 R: 0.8497 F1: 0.8101	84.89	P: 0.8034 R: 0.8489 F1: 0.8052
7	RNN	84.33	P: 0.7112 R: 0.8433 F1: 0.7716	84.24	P: 0.7097 R: 0.8424 F1: 0.7704

Random Forest ranked second among all evaluated models, achieving an accuracy exceeding 82.9% in both experiments. Its ensemble structure enables robust classification by aggregating predictions from multiple decision trees, thereby reducing overfitting and improving generalization.

Among the traditional statistical approaches, Logistic Regression and Naïve Bayes produced lower accuracies of approximately 64% and 67%, respectively. Although these models exhibited relatively high precision values above 0.83, their recall values were considerably lower, suggesting limitations in identifying positive diabetes cases compared with ensemble-based methods.

ANN and CNN achieved the highest classification accuracies among all evaluated models, demonstrating that deep learning architectures can effectively learn complex patterns from tabular health indicators. However, XGBoost achieved a higher F1-score, indicating a better balance between precision and recall. This suggests that while deep learning models provide excellent overall accuracy, ensemble learning methods remain highly competitive for healthcare classification tasks.

C. Discussion

The experimental results highlight three important findings. First, the application of SMOTE successfully mitigated class imbalance, enabling all the 4 machine learning models to achieve stable classification performance. Second, ensemble learning techniques, particularly XGBoost and Random Forest, substantially outperformed conventional machine learning approaches. However, ANN and CNN achieved slightly higher classification accuracies, indicating that both deep learning and ensemble learning methods are effective for diabetes prediction. Third, feature-removal analysis revealed that diabetes prediction performance is largely preserved despite the exclusion of one feature, demonstrating the robustness of the predictive framework.

A notable observation is that deep learning models achieved competitive performance despite being trained on the original imbalanced dataset, whereas machine learning models required SMOTE balancing to achieve comparable results. This finding highlights the ability of deep learning architectures to learn discriminative representations on the BRFS 2015 dataset. However, the observed performance may vary when applied to datasets with different population characteristics, feature distributions, or prevalence rates.

V. CONCLUSION AND FUTURE WORKS

This study conducted a comparative evaluation of four machine learning models (Logistic Regression, Naïve Bayes, Random Forest, and XGBoost) and three deep learning models (ANN, CNN, and RNN) for diabetes prediction using the BRFS 2015 dataset. Rather than proposing a novel prediction architecture, the study aimed to assess the relative performance of established methods under consistent preprocessing and evaluation conditions.

The experimental results demonstrated that ANN achieved the highest accuracy of 84.99%, while XGBoost achieved the highest F1-score among the machine learning models.

The results indicate that both deep learning and ensemble learning approaches are effective for diabetes prediction on the BRFS 2015 dataset. However, caution should be exercised when generalizing these findings to other populations or healthcare datasets because model performance may depend on dataset characteristics and data quality. Furthermore, the feature-removal experiments resulted in only minor performance variations across all machine learning models, indicating that the removed feature contributed limited predictive information and that the remaining features were sufficient for effective diabetes classification.

A limitation of this study is that the evaluation was conducted using a single public dataset. Although the BRFS dataset is large and widely used, the findings may not fully represent performance on datasets collected from different populations, healthcare systems, or clinical environments. External validation on independent datasets is required before broader conclusions can be drawn regarding model generalizability.

Future research may focus on incorporating advanced feature selection and explainable artificial intelligence techniques, such as SHAP and LIME, to better understand the contribution of individual health indicators to diabetes prediction. Hyperparameter optimization methods, including Bayesian Optimization and Genetic Algorithms, could also be explored to further improve model performance. In addition, hybrid deep learning architectures that combine CNN and RNN components may be investigated to capture more complex relationships within health data. The proposed framework can be further validated using larger and more diverse clinical datasets from different populations to enhance its generalizability and practical applicability in real-world healthcare environments. Finally, integrating the prediction framework into intelligent clinical decision support systems could assist healthcare professionals in early diabetes screening and risk assessment. Future studies should also investigate the application of class-weighting techniques and focal loss functions in deep learning models to address class imbalance without relying on synthetic oversampling. Such approaches may further improve minority-class recognition while preserving the original data distribution.

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